Coordination Compounds

SHORT NOTES

COORDINATION COMPOUND

A compound containing coordinate bonds, typically between a central metal atom and a number of other atoms or groups.

WERNER THEORY OF COORDINATION COMPOUNDS

According to Werner (father of Co-ordination Chemistry) transition metals possess two types of valencies:

Primary Valency (Ionisable Valency)

1. It is satisfied by only anions.
2. It is referred to as oxidation state.

Secondary Valency (Non-Ionisable Valency)

1. It is satisfied by anions or neutral molecules or rarely with cations.
2. The groups satisfying secondary valencies are called ligands.
3. The number of secondary valencies is called coordination number.
4. The ligands are directed in space around the central metal atom in different ways. This leads to a definite geometry to the molecule.

Difference Between Double Salt and Complex Salt

Double salts	Coordination Compound (complex salt)
1. Double salts exist only in the solid state.	1. Coordination compound retain in both the form solid as well as liquid.
2. In aqueous solution, they dissociate completely into ions.	2. They will remain as a compound in aqueous state and does not dissociates into ions.
E.g.,$KCl.MgCl_2.6H_2O$ (Carnallite)	E.g., $K_4[Fe(CN)_6]$, $K_3[Fe(CN)_6]$, $Na_3[Ag(S_2O_3)_2]$

DEFINITIONS OF SOME IMPORTANT TERMS IN COORDINATION COMPOUNDS

Coordination entity or coordination sphere

(*i*) A coordination entity constitutes a central metal atom or ion bonded to a fixed number of ions or molecules.

(*ii*) It is enclosed in square bracket and is collectively termed as the **coordination sphere**.

(*iii*) The ionisable groups are written outside the bracket and are called **counter ions.**

Central Atom/Ion

(*i*) In a coordination entity, the atom/ion to which a fixed number of ions/groups are bound in a definite geometrical arrangement around it, is called the central atom or ion.

Ligands

(*i*) Species which donate lone pair/ electron pair is called as ligand. on the basis of the number of electron pairs available for donation; ligands are classified as:

Ligands

- On the basis of charge
  - Neutral
  - Anionic
  - Cationic
- On the basis of denticity
  - Monodentate
  - Bidentate
  - Polydentate
  - Flexidentate
  - Ambidentate

Coordination Number

(*i*) The coordination number (C.N) of a metal ion in a complex can be defined as the number of ligand donor atoms to which the metal is directly bonded.

Oxidation Number of Central Atom

(*i*) The oxidation number of the central atom in a complex is defined as the charge it would carry if all the ligands are removed along with the electron pairs that are shared with the central atom.

(*ii*) The oxidation number is represented by a roman numeral in parenthesis following the name of the coordination entity.

Homoleptic and Heteroleptic Complex Compound

Homoleptic complex:

(*i*) The complex in which the central metal atom (or) ion bound with only one kind of ligand.

Example: $[Fe(CN)_6]^{-4}$, $[Co(H_2O)_6]^{+3}$

Heteroleptic complex:

(*i*) The complex in which the central metal atom (or) ion bound with more than one kind of ligand.

Example: $[CO(NH_3)_5Cl]^{+2}$

RULES FOR NAMING COORDINATION COMPOUNDS

- The anion is named after the cation.
- The complex part is written as one word and name of coordination compounds is started with a small letter.
- In naming coordination sphere, ligand are named first in alphabetical order followed by metal atom and then oxidation state of metal by roman numeral in parenthesis.
- Name of anionic ligand ends in - O.
- Name of cationic ligand ends in - ium.
- The prefixes di-, tri- tetra-, penta- and hexa- are used to indicate the number of each ligand. If the ligand name includes such a prefix, the ligand name should placed in parentheses and preceded by bis-(2), tris (3), tetrakis (4), penta kis(5), hexakis(6)
  $[Cr(NH_3)_3(H_2O)_3]Cl_3$ is named as:
  triamminetriaquachromium(III) chloride
  $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$ is named as:
  tris(ethane-1, 2-diammine)cobalt(III) sulphate

ISOMERISM IN COORDINATION COMPOUND

Structural Isomerism

- This isomerism arises due to the difference in the structures of complexes. It is of four types:
  - (***a***) **Ionisation isomerism:** Same molecular formula but gives different ionisable species. (Only anionic)
  - (***b***) **Hydrate isomerism:** Same molecular formula but different number of water molecules associated with central metal.
  - (***c***) **Linkage isomerism:** Structural isomerism shown by ambidentable ligands. e.g., $[Fe(NH_3)_5(SCN)^{2+}]$ $[Fe(NH_3)_5(NCS)^{2+}]$
  - (***d***) **Coordination isomerism:** Isomers having both anion and cation as complex entity. Can inter change position of ligands as well as metal. e.g., $[Cr(NH_3)_6][Co(CN)_6]$ $[Co(NH_3)_6][Cr(CN)_6]$

Stereo Isomerism

(*i*) Geometrical Isomerism

- Tetrahedral complexes cannot exhibit geometrical isomerism.
- Square planar complexes of the type Ma_4, Ma_3b, Mab_3 do not exhibit geometrical isomerism.
- Square planar complexes of the type Ma_2b_2, Ma_2bc and M_{abcd} exhibit geometrical isomerism.

  Example: $[Pt(NH_3)_2Cl_2]$

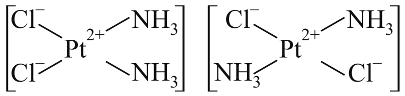

Fig.: Cis-isomer & Trans-isomer of $[Pt(NH_3)_2Cl_2]$

- Mabcd type of complexes will have 3 geometrical isomers i.e. 2 cis and 1 trans isomer
- Square planar complex having unsymmetrical bidentate chelating ligands with general formula $[M(AB_2)^{\pm n}]$ exhibit geometrical isomerism
- Octahedral complexes of the type Ma_4b_2, Ma_4bc and Ma_3b_3 complexes exhibit geometrical isomerism.

  Example: $[CoCl_2(NH_3)_4]^+$

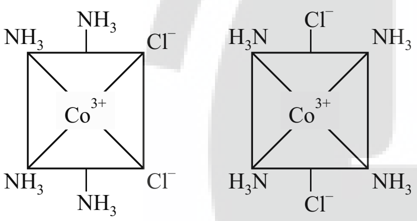

Fig.: Cis-isomer & Trans-isomer of $[CoCl_2(NH_3)_4]^+$

- Ma_3b_3 type of complexes exhibits meridian and facial type of isomers

(*ii*) Optical Isomerism

- Coordination number four, tetrahedral complexes of the type (M_{abcd}) exhibit optical isomerism.

  Example: $[Pt(NH_3)(H_2O)Cl(NO_2)]$

- Optical isomers are also generally provided by the octahedral coordination compounds having bidentate (or) polydentate ligands.

  Example: $[CoCl_2(en)_2]^+$, $[Cr(C_2O_4)_3]^{2-}$

  $[Cr(NH_3)_2Cl_2en]^+$; $[PtCl_2(en)_2]^{2+}$

  Here 'en' represents $H_2NCH_2CH_2NH_2$

- Optical isomerism is shown by the complexes having chiral structures. They rotate the plane of polarised light in opposite directions.

  Complexes with C.N. = 4

- Octahedral complexes containing polydentate ligands are optically active.

  Example: $[Co^{III}(EDTA)^-]$.

Effective Atomic Number (EAN)

- EAN = Z_{metal} – oxidation state of the metal + 2 × coordination number of the metal.
- Ex. (1) In $K_4[Fe(CN)_6]$ complex

  EAN of Fe = 26 – 2 + 6 × 2 = 36.

BONDING IN COORDINATION COMPOUNDS

Valence Bond Theory

Postulates

- The vacant orbitals of central metal atom hybridise and overlap with the orbitals of the ligands each containing lone pair of electrons.
- A ligand contains atleast one lone pair of electrons which can be used for bonding with the central metal ion.
- The electrons in the metal orbitals may undergo regrouping even **against Hund's rule**.
- Ligand orbitals overlap the vacant metal orbitals to from a strong coordinate covalent bond to the extent possible.
- **sp^3 hybridisation:** The structure of the complex is **tetrahedral.** $[Ni(CO)_4]$,$[NiCl_4]^{2-}$
- **dsp^2 hybridisation:** The Structure of the complex is **square planar**.

  Ex: $[Cu(NH_3)_4]^{2+}$, $[Cu(CN_4)^{2-}]$, $[Ni(CN)_4]^{2-}$

- ***sp^3d*** hybridisation: The Structure of the complex is **trigonal bipyramidal**.

  Ex: $[Fe(CO)_5]$,$[CuCl_5]^{3-}$

Inner and Outer Orbital Octahedral Complexes

- In an **inner orbital complex**, inner $(n–1)d$ orbitals and the outer *ns* and *np* orbitals hybridise.

  Ex: $K_4[Fe(CN)_6]$. The hybridisation involved is d^2sp^3.

- Inner orbital complexes are also called **spin paired** or **low spin** or **strong field or covalent complexes**.
- In an **outer orbital complex**, outer *ns*, *np* and *nd*-orbitals hybridise.

  Ex: $[CoF_6]^{3-}$ ion, $[Co(H_2O)_6]$, $[Mn(H_2O)_6]^{2+}$, $[Cr(NH_3)_6]^{2+}$

- The hybridisation involved is ***sp^3d^2***.
- Outer orbital complexes are also called **spin free or high spin** or weak field or **low field or ionic complexes.**

Magnetic Properties of Coordination Compounds

- The magnetic moment of the complexes can be easily calculated using the formula
- $\mu = \sqrt{n(n+2)}$B.M

  (n = number of unpaired electrons)

- Thus, for d^4, d^5 and d^6 cases, two vacant *d*-orbitals are only available for hybridisation as a result of pairing of 3*d* electrons which leaves two, one and zero unpaired electrons respectively.

$[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[Co(C_2O_4)_2]^{3-}$ are inner orbital complexes involving d^2sp^3 hybridisation, the former two are paramagnetic and the latter diamagnetic. $[MnCl_6]^{3-}$, $[FeF_6]^{3-}$ and $[CoF_6]^{3-}$ are outer orbital complexes involving sp^3d^2 hybridisation and are paramagnetic having four, five and four electrons respectively.

Crystal Field Theory (CFT)

Crystal Field Splitting of *d*-orbitals

- The other two *d*-orbitals i.e.,$d_{x^2-y^2}$ and d_{z^2} which are oriented along the axes are called **e_g-orbitals**.
- This spliting of *d*-orbitals of metal ion under the influence of approaching ligands is called **crystal field splitting**. It is designated by Δ and is called crystal field splitting energy.

(*a*) Crystal field splitting in octahedral complexes

- The energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of the three t_{2g} will decrease by $(2/5)\Delta_o$. Consequently. In the **e_g set of orbitals has higher energy than t_{2g} orbitals**.

(*b*) Crystal field splitting in tetrahedral complexes

- **In this situation, the t_{2g} set of orbital** lie relatively nearer to the approaching ligands and therefore **t_{2g} set of *d*-orbitals have higher energy than e_g set of orbitals**.
- Relationship between Δ_t and Δ_o is given as ; $\Delta_t = \frac{4}{9}\Delta_O$
- While entering electrons into *d*-orbitals after splitting in ligand field, Hunds and pauli's principle to be followed.

Factors Determining the Magnitude of the Orbital Splitting Energy

The following factors influence the magnitude of Δ.

(1) Nature of ligand

- The ligands which cause only small Δ value are called **weak field ligands** while those which cause a large Δ value are called **strong field ligands.**
- The ligands can be arranged in the increasing field strength in spectrochemical series.

$$\underbrace{I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O}_{\text{weak ligand}}$$

$$< \underbrace{NCS^- < edta^{4-} < NH_3 < en < NO_2^- < CN^- < CO}_{\text{strong ligand}}$$

Such a series is termed as spectrochemical series.

For d^4 configuration, the two possibilites are:

(*i*) If $\Delta_o < P$, the fourth electron enters one of the e_g orbitals giving the configuration $t^3_{2g}e_g^1$. Ligands for which $\Delta_o < P$ are known as **weak field ligands** and form high spin complexes.

(*ii*) If $\Delta_o > P$, it becomes more energetically favourable for the fourth electron to occupy a t_{2g} orbital with configuration $t_{2g}^4e_g^0$. Ligands which produce this effect are known as **strong field ligands** and form low spin complexes.

(2) Oxidation state of the metal ion

- Higher the ionic charge on the central metal ion, the greater the value of Δ.
- The metal ion with higher oxidation state causes **larger Δ** than is done by the ion with lower oxidation state.

(3) Nature of the metal ion

- Coordination entities of second and third transition series have a greater tendency to form low spin complexes as compared to the first transition series.

Colour in Coordination Compounds

- In octahedral complexes Δ vary from one metal ion to another and the nature of the ligands.
- For example, three complexes of Co^{3+} as $[Co(H_2O)_6]^{3+}$, $[Co(NH_3)_6]^{3+}$ and $[Co(CN)_6]^{3-}$. **According to spectrochemical** series, the crystal field **splitting energies** will be in the order of **ligands as $H_2O < NH_3 < CN^-$**

BONDING IN METAL CARBONYLS

- The M–C σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of the metal.
- The M–C π bond is formed by the donation of a pair of electrons from a filled d-orbital of metal into the vacant anti-bonding π* orbital of carbon monoxide.
- The metal to ligand bonding creates a **synergic effect** which strengthens the bond between CO and the metal.

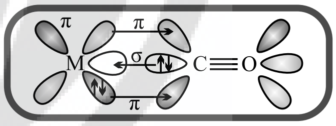

Fig.: Synergic bonding

Structure of Metal Carbonyls

- Tetracarbonyl nickel (0) is tetrahedral.
- Some carbonyls have metal-metal bonds.
- Decacarbonyl dimanganese (0) is made up of two square pyramidal $Mn(CO)_5$ units joined by Mn-Mn bond.

APPLICATION OF COMPLEXES

- Fe(III) ion in group III is identified as $[Fe(NCS)_6]^{3-}$ ion which has blood red colour.
- In the electroplating of Ag, large amounts of silver are held as the complex, $K[Ag(CN)_2]$.
- Haemoglobin is a complex which contains Fe^{+2} ions.
- Chlorophyll is a complex which contains Mg^{+2} ions.
- Vitamin - B_{12} is a complex which contains Co^{3+} ions.